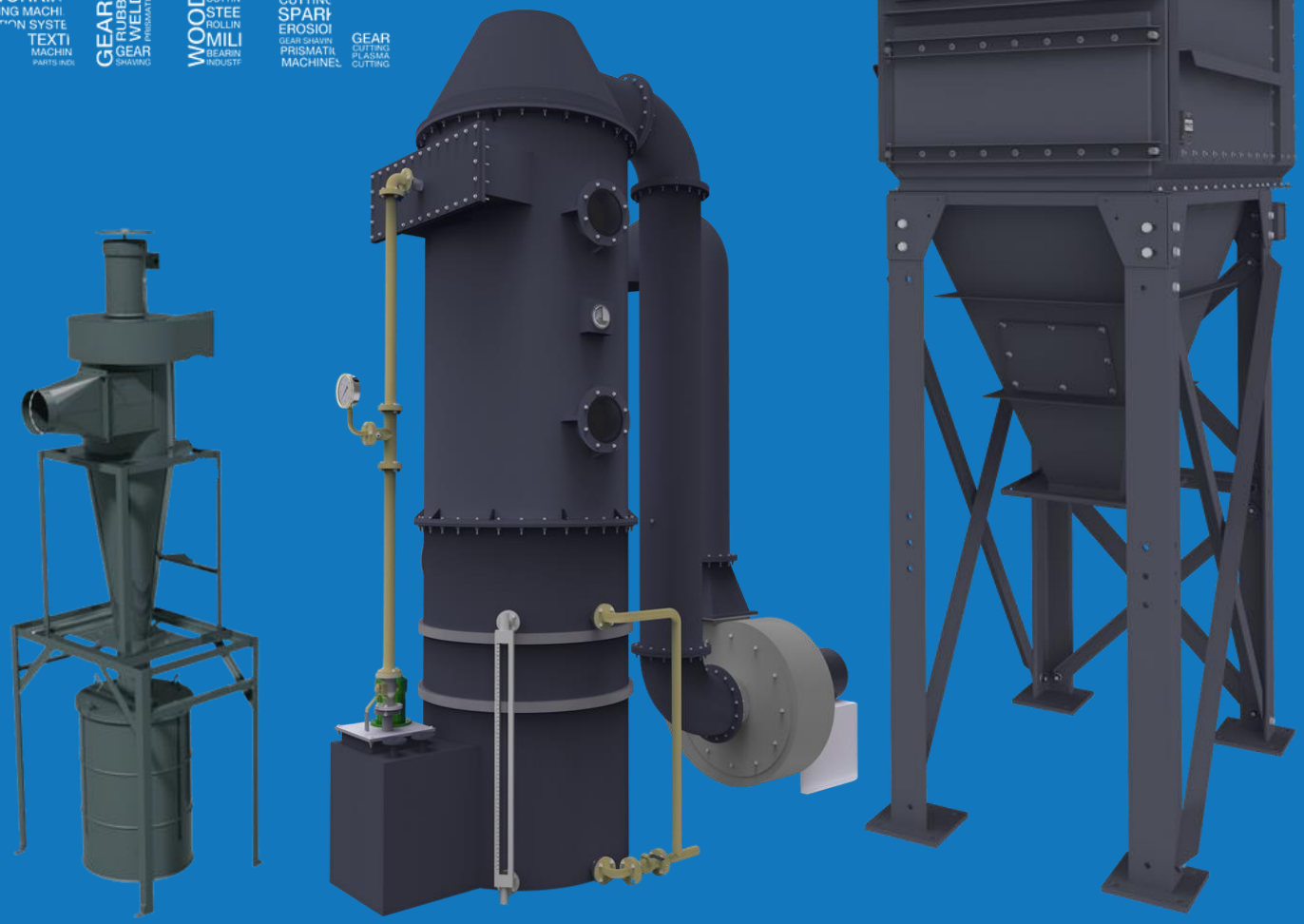
DUST MIST FUME SMOKE





- AIR POLLUTION CONTROL SYSTEMS
- DE DUSTING SYSTEMS
- CENTRIFUGAL FANS & BLOWERS
- PNEUMATIC CONVEYING SYSTEMS
- COMPONENTS RELATED TO ABOVE SYSTEMS

PRODUCTS

- Air Pollution Control Systems
- De Dusting Systems
- Centrifugal Fans & Blowers
- Pneumatic Conveying Systems
- Cyclone Mechanical Dust Collector
- Unitary Dust Collectors/ Air Filters Manual/Motorized Shaker Dust collector
- Scrubber (Cyclonic spray scrubbers & Venturi Scrubber)
- Roof Extractors / Roof Ventilators Rotary Air lock Valves
- Screw Conveyers
- Venturi Ejectors / Air Conveyor Nozzles Air Washers
- Air Handling Units & Central Humidification and Air Cooling plant

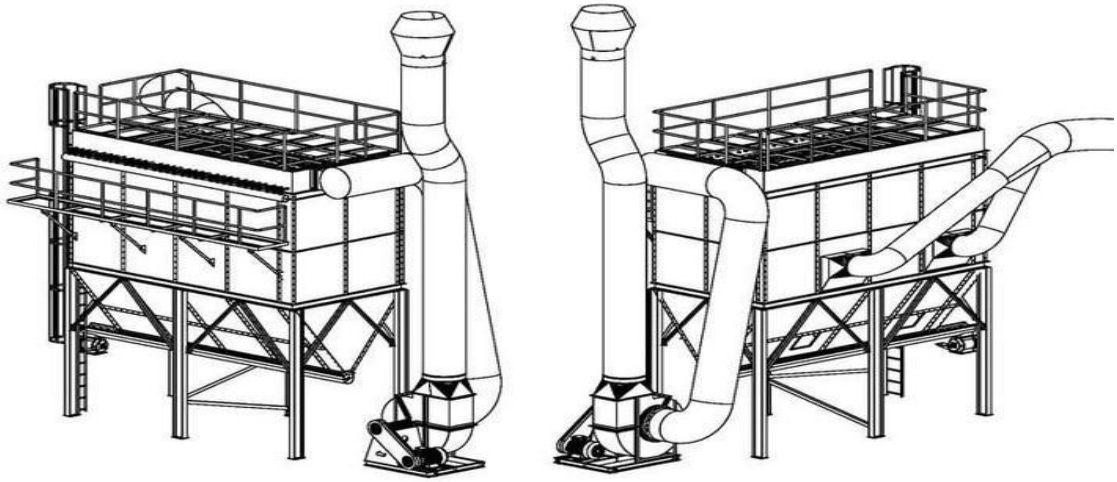
Experts in Custom Air Pollution Control Systems

Since 2020, we are providing air pollution control equipment system with advanced design and we deliver tailored dust collection systems for diverse industrial applications.

Our durable and cost-effective solutions ensure optimal performance while addressing environmental challenges. We are Trusted supplier for innovative and reliable air pollution control systems.

Comprehensive Air Pollution Control Solutions

We supply full range of air pollution control equipment, accessories, and turnkey systems. Our expertise spans industrial air pollution control, dust management, pneumatic conveying, and custom solutions tailored to diverse industry needs.



INDUSTRIAL APPLICATIONS

- Cement Plants
- Steel Plants
- Power Plants
- Incinerators
- Textile
- Industry Sugar Plants
- Paper Mills
- Coal Processing
- Fly Ash Management
- Pharmaceutical Sector
- Woodworking Material
- Recovery ...and many other industrial applications.

CYCLONE SEPARATOR

High Efficiency Medium Flow Type Cyclones :

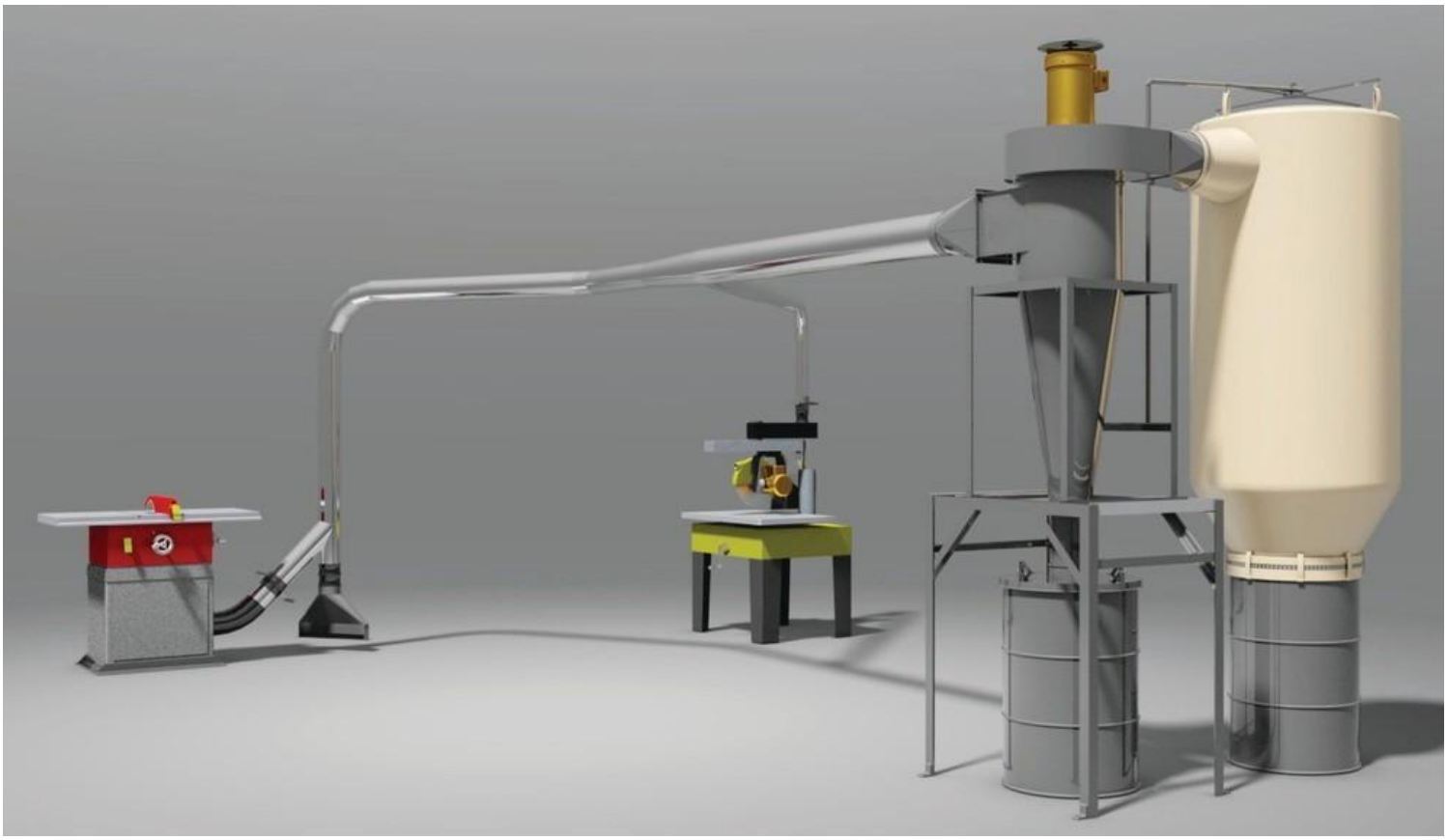
We are offering a highly popular range of mechanical dust separators & collectors generally termed as Cyclones Due to centrifugal separation, spherical & short fibrous or cylindrical dust particles from the dusty gas flows, are thrown to the periphery & thence to the bottom part Cleaned gas / air leaves thro' central discharg e. Tangential entry at volute top provides high efficiently swirling & allows higher flow.

Twin, triple or Quadruple cyclone group models are offered for higher flow at higher efficiencies. Construction & material are to suit duty & temperatures.

Accessories like Rotary air-lock valve & Exhauster fans are offered from standard range. Recommended: for: Boilers, Dust control in foundries chemicals, plastics, pharmaceuticals, Food & feed, crushing, conveying, screening, grinding & similar operations.

Capacity : 1000 CMH To 45,000 CMH
Pressure Drops UP to 120 mm WC

Dust Collection efficiency up to 95% on Particulate larger than 10 microns.



Medium Efficiency High Flow Type Cyclones:

We are offering a highly popular range of mechanical dust separators & collectors generally termed as Cyclones Due to centrifugal separation, spherical & short fibrous or cylindrical dust particles from the dusty gas flows, are thrown to the periphery & thence to bottom part Cleaned gas / air leaves thro' central discharge . Tangential entry at Helical top provides smooth turbulence free dust descent & separation. (High temperature constructions optional)

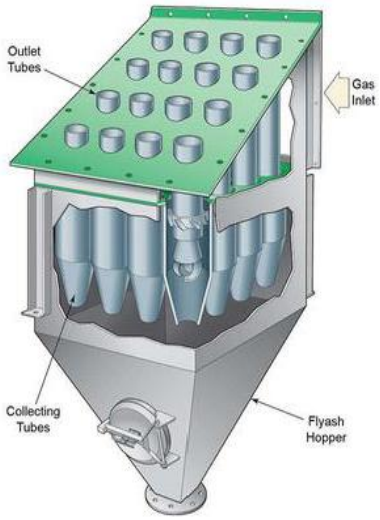
MECHANICAL DUST COLLECTOR

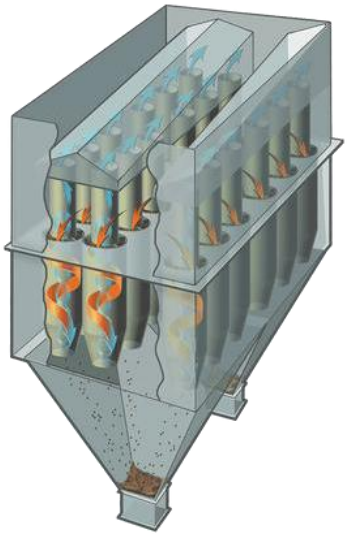
A parallel bank of high efficiency small diameter cyclone tubes are arranged to achieve dust/particulate collection efficiencies up to 95% on 5 micron size .If required for high temperature & high moisture conditions Multiclone are specially constructed & offered. Accessories like Rotary air-lock valves, screw conveyer or ID fans are made available in standard range. Recommended for : Air pollution control in cement/Bauxite processing ,Boilers ,kilns, shot blasting ,cupolas ,fine dust control on application involving high temperature high moisture &large flow

Capacity :3000 To 100,000 CMH
Pressure Drops UP to 150 mm WC

Dust Collection efficiency up to 95% on 5 micron size.







PULSE JET CARTRIDGE TYPE FILTER

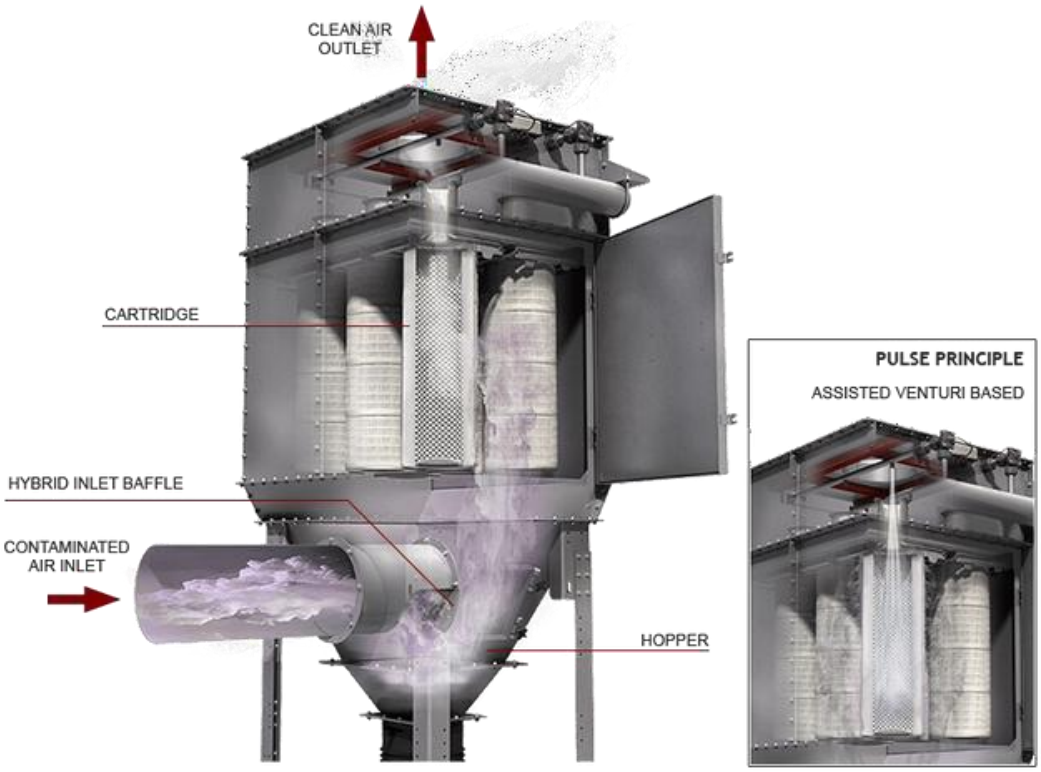
Cartridge type dust-collector is mainly used for the collection of suspended dust and particulates, both for air pollution control and, as part of the production process, to ensure efficient and continuous on-line dust collection. The machine has the advantages of stable operation, large air... Cartridge type dust-collector is mainly used for the collection of suspended dust and particulates, both for air pollution control and, as part of the production process, to ensure efficient and continuous on-line dust collection. The machine has the advantages of stable operation, large air volume, small size and convenient maintenance. Can be widely used in metallurgy, machinery, casting, chemical and other departments, dust concentration between 20 ~ 80mg / m3.

FEATURES

- Low height front loading of cartridge removal & insertion.
- Continuous Duty operation.
- Lower Pressure drop in Comparison with hose bag type bag filter
- Supplied in Ready to start condition

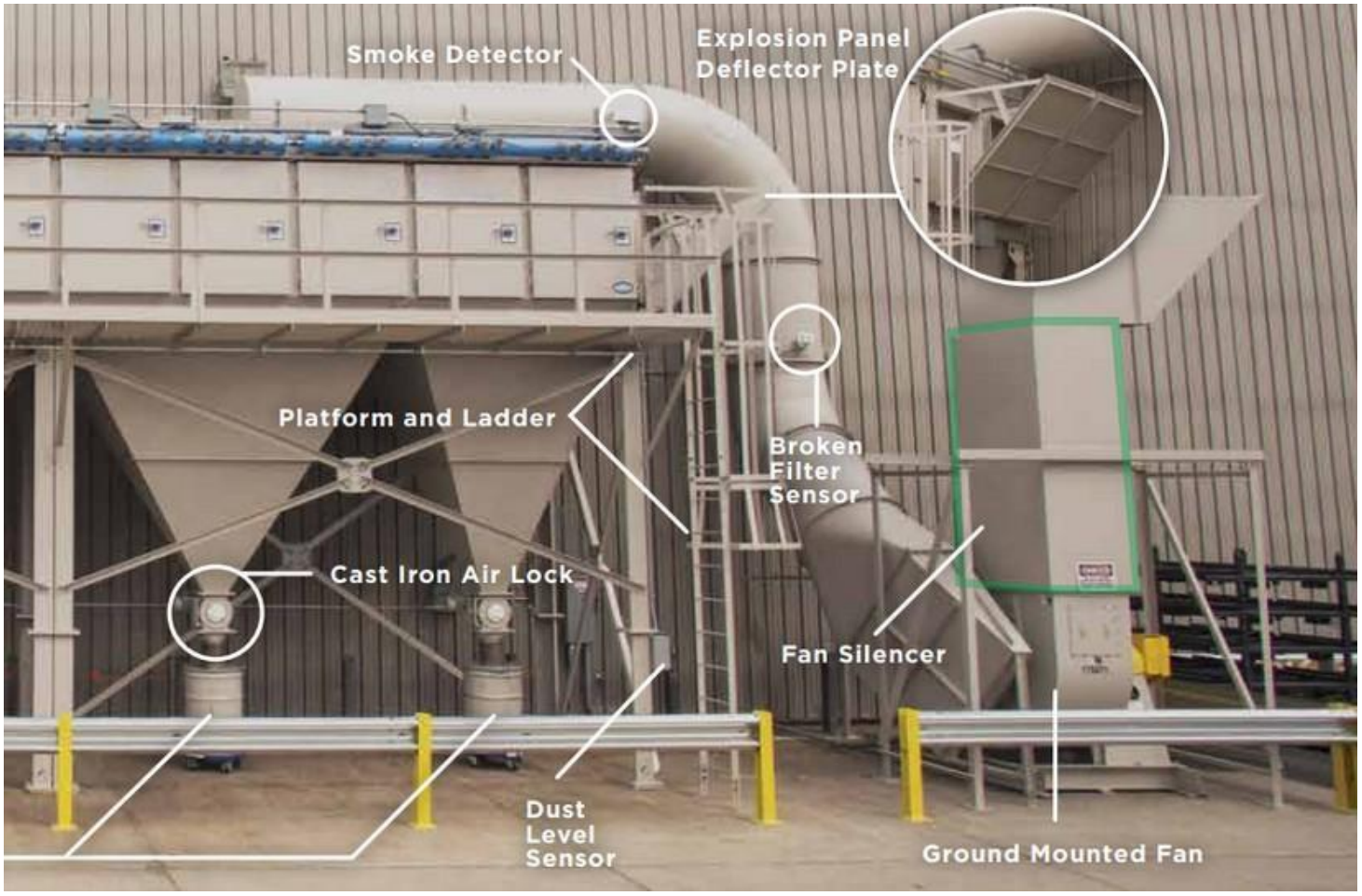
Application / uses

- Sand Blasting
- Shot Blasting
- Polymers
- Ceramics
- Grinding & Polishing
- Abrasive Mfg.
- Food Processing
- Spices Pneumatic conveying systems
- welding & cutting mineral Grinding Pigments & Chemicals



How it Works

Dust laden air is pulled into the collector via an induced draft. The air travels through our engineered diffusion-type baffle located at the inlet of the collector. The air is then pulled through a pleated cartridge filter, where dust is captured on the outer surface of the filter. Clean air passes through the filter to the clean air outlet. The cartridges are automatically cleaned of the collected dust as the pressure increases. The on-line cleaning system consists of a compressed air pulse initiated by a timer and activated through a solenoid valve. The pulse system has a pressure flow characteristic to give maximum power in a minimum time frame. This stops air flow through the element in the filtering mode and reverses the flow in order to dislodge the collected dust.



PULSE JET HOSE BAG TYPE FILTER



Pulse Jet Hose Bag type filters are known as conventional filters. Bag filters are filters used in industrial units for filtering dusty gases. The basic function of bag filter is to separate dust particles for dusty gases released by different production processes. The structure of a bag filter is made up of a fabric bag crafted using different kind of fabrics and a motor that powers the machine. The mechanism of bag filters works in a very smooth and convenient way.
Bag filters are used in all industrial small or large units to collect dust particles blown out in toxic release.

FEATURES

- Modular Designs available starting from 1600 m3/hr to 400000 m3/hr
- Tool less Installation/replacement of bag & Cages.
- Can Handle up to 300°C Hot gas.
- Easy to install & commissioning.

APPLICATION/USES

- Foundry & Steel
- Cement
- Boilers Flue Gas
- Kiln Exhaust
- Solid drying
- Mining and minerals
- Wood-Working and laminates
- Hot Gases and fumes Coal
- Handling
- Glass Particle
- Board Mfg.



PULSE JET ENVELOPE TYPE FILTER

Envelope filters is designed to cater for the most common dust problems, which usually involve a ducted system. Larger systems utilise a number of units in a modular form, or other types of filter from our range.



FEATURES

- Downdraft Airflow Direction for best dust separation at inlet
- Reliable & Efficient for critical powder/Dust Best for Product Recovery Continuous Duty Operation
- Lower Pressure Drop In Comparison with hose bag type bag filter

APPLICATION / USES

- Rubber Processing
- Master batch and Pigments
- Powder recovery
- Metal recovery
- Food processing
- Foundry & steel

Unitary Dust Collectors/ Air Filters



Model:
Manual Shaker Type
Motorized Shaker type
Automatic Pulse Jet type.

Dust collectors are designed for Powerful Suction & High Air Extraction, No Maintenance, Easy Cleanability, Compact size and most effective & widely employed, Versatile (Automatic) Useful single utility for multiple uses.

Readily Applicable for :

Air Pollution Control, Duct Capture & Control, Powder Product Recovery Wood working operations, Grinding, turning, milling, sanding, buffing, polishing, crushing pulverizing powder mixing chemical Pharmaceutical paints plastic ceramics Glass Food Processing Cosmetics Engineering Fibrous processes ETC.

Capacity: 500 To 4500 CMH
Static Pressure: 50 mm to 250 mm WC
Dust Collection efficiency: As per model selection.

WET SCRUBBER



Wet scrubbers work by bringing polluted gas streams into contact with a scrubbing liquid, usually water or a water-based solution, to remove contaminants such as particulate matter and harmful gases. The key steps in their operation are:

Gas Introduction: Polluted gas enters the scrubber chamber, typically from the bottom or side.

Liquid Contact: The scrubbing liquid is sprayed or introduced into the gas stream as a fine mist or through a packed bed, maximizing contact between gas and liquid.

Pollutant Capture: Particulate matter impacts and sticks to liquid droplets, while gaseous pollutants dissolve or chemically react with the liquid.

Separation: The pollutant-laden droplets coalesce and fall into a collection chamber or sump.

Clean Gas Exit: The cleaned gas, now free of pollutants, exits the scrubber, often passing through a mist eliminator to remove excess moisture.

Liquid Treatment: The scrubbing liquid containing pollutants is collected for treatment, recycling, or disposal.



Packed Tower (Packed Bed) Wet Scrubbers

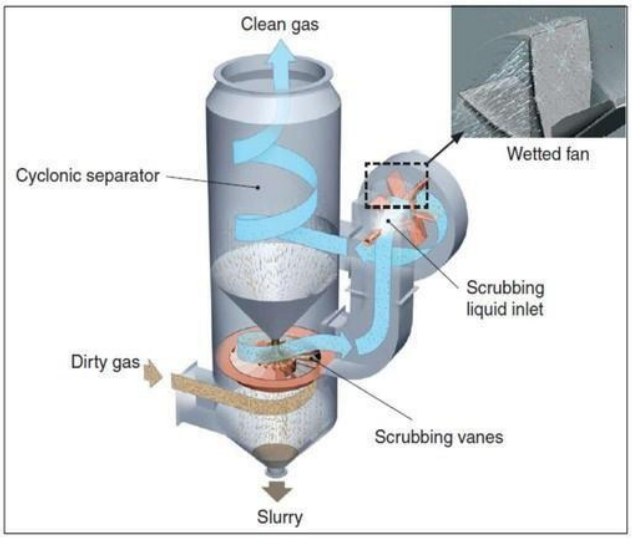
- Contain packing material to increase contact between scrubbing liquid and gas.
- Primarily used for gas-phase pollutant removal like sulfur dioxide and odors.
- High efficiency (90–99%) for gas absorption but less effective for particulate matter due to clogging risk.

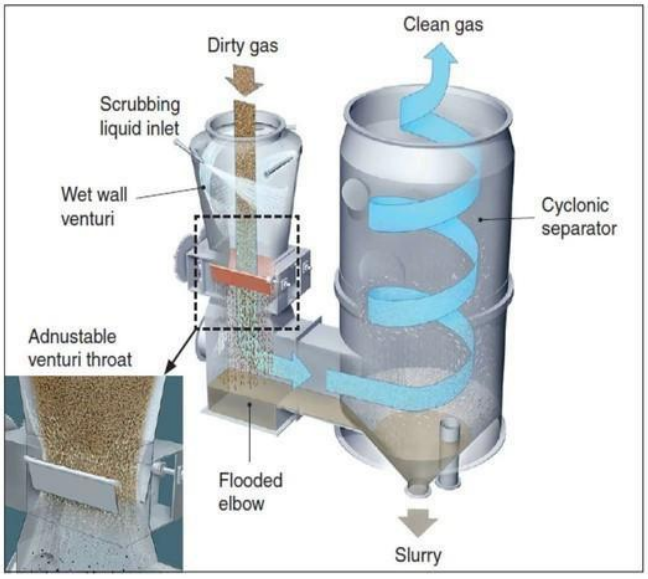
Cyclonic spray scrubbers

Cyclonic spray scrubbers are used to help in air pollution control and they use the features of a spray chamber and a dry cyclone to remove pollutants from gas streams.

The inlet gas enters the chamber tangentially, swirls through in a corkscrew motion, and exits while liquid is sprayed inside the chamber.

As the gas swirls around the chamber, pollutants collide with the liquid and are removed by the droplets as they are thrown to the walls, and washed back down and out.





Venturi scrubber

Venturi scrubber consists of a converging section, a throat (the narrowest part of the venture tube) and a diffuser.

The dust/gas mix flows through the Venturi tube and reaches top speed in the throat section. Thereafter, the mixture passes into the diffuser where the speed drops again.

Liquid is added to the gas flow either in the throat section or prior to it. Intensive mixing takes place between the gas and the liquid in the throat section of the venture tube. Due to the high speed realized by the gas and liquid, water is released in fine water droplets.

POPPET DAMPER

This unusual design feature a pneumatic or linear electric driven shaft operating vertically rising horizontal blade. Steel plant designers favor this construction for their very high reliability under exacting conditions of high temperature and dust load.

Typical feature include quick acting pneumatic cylinder operation and an accumulator driven quick shut – off under pneumatic air supply failure condition.
It incorporates a simplex leaf – seal arrangement for maintaining sticky flue gas particulates above their condensation temperature.
Poppet valve employ a seal disc that is capable of extremely rapid and used in critical gas ducting applications that require rapid diversion or isolation of gases.



Application

Isolation of a single duct (Two – Way)
Control flow from one gas path to either of two new paths (Three-way)
Control flow from two gas paths to either of two new paths (Four – way)
Very quick shut-off action approx. one second or less cycle time

WELDING FUME EXTRACTOR



We offers comprehensive, large-scale welding fume extraction systems designed to efficiently capture and remove hazardous welding fumes from multiple welding stations simultaneously, ideal for large workshops, industrial plants, and training facilities. Our centralized extraction units use powerful fans and multi-stage filtration to capture up to 99.97% of harmful fumes, including fine dust, heavy metals, and toxic gases, directly at the source through flexible extraction arms or integrated torch extraction

Key advantages include:

- Centralized system design for easy management of multiple welding stations with a single, high-capacity collector.
- Energy-efficient operation with variable frequency drives that reduce power consumption and noise during idle times.
- Robust, low-maintenance construction suitable for continuous industrial use.
- Flexible extraction arms with long reach (up to 20 feet) to cover large work areas without interfering with welding tasks.
- Automatic or manual filter cleaning options to ensure consistent performance and reduce downtime.

DOUBLE BAG DUST COLLECTOR



The Double Bag Dust Collector a robust and efficient dust extraction system designed for industrial environments such as woodworking shops, metalworking facilities, and manufacturing units. Engineered for high-volume dust collection, this unit features a powerful motor and dual filtration bags to ensure superior air quality and a cleaner workspace.

Key Features

- High-Efficiency Dual Bag System:
- Equipped with two large-capacity filter bags, this collector maximizes dust and particulate capture, reducing airborne contaminants and maintaining a healthier work environment.
- Powerful Suction Performance:
- The heavy-duty blower delivers strong airflow, efficiently extracting dust and debris from multiple machines or workstations simultaneously.
- Durable Construction:
- Built with industrial-grade materials for long-lasting performance and reliability in demanding applications.
- Easy Maintenance:
- Quick-release bag clamps and accessible design make bag replacement and routine maintenance simple and hassle-free.
- Compact Footprint:
- Space-saving design allows easy integration into various workshop layouts without compromising on extraction power.

BENCH DUST COLLECTOR



The Bench Dust Collector is a high-performance, compact extraction unit designed to provide efficient dust and fume removal directly at the work surface. Ideal for applications such as grinding, welding, sanding, and polishing, this bench-mounted system ensures a cleaner, safer, and more productive workspace.

Applications

- Welding and metalworking
- Grinding and sanding
- Soldering and polishing
- Laboratory and educational settings

ROOF EXTRACTORS / ROOF VENTILATORS



Roof-mounted fans eliminate the harmful effects created when high temperatures and humidity make work and physical effort difficult. They also help extend the life of the structural elements of buildings and warehouses by preventing condensation and humidity. An adequate ventilation system in industrial buildings will considerable reduces cooling and heating expenditure. This type of system easily, continuously and effectively eliminates heat and humidity.

Capacity: 1000 CMH to 5000 CMH Size : 450 mm to 1200 mm

ROTARY AIR LOCK VALVES



Rotary Airlock Valves are a type of rotary feeder and are often referred to simply as a Rotary Airlock or Rotary Valve.
They provide feeding and metering functionality of a standard rotary feeder but their primary duty is to maintain pressure and provide an optimal airlock seal between different product conveying transitions, such as gravity to pneumatic conveying lines.
An optimal airlock condition is essential to prevent air pressure loss in high pressure and high temperature conveying systems.

We can supply Rotary Air Lock feeders from 1 TPH to 5 TPH capacity and inlet sizes ranging from 150 mm to 450 mm in round shape.

SCREW CONVEYERS



Screw Conveyor is mainly used for conveying the materials in various industries. We make the conveyors in different sizes and as per customer's requirements. Screw conveyors made in SS & MS.

Material Handling Capacity : 50 kg/Feet to 200 Kg/Feet

Length : 2 Feet to 20 Feet

VENTURI EJECTORS / AIR CONVEYOR NOZZLES

Venturi Ejector for use in transporting powders, pellets, and bulk solids, typically using motive pressures. We provides complete Venturi transport systems, including blowers, valves, receivers, etc.

We make the nozzles in different sizes and as per

customer's requirements. Venturi ejector made in SS & MS.



SPOT HUMIDIFIER

We Offer high capacity evaporation, effective mist dispersal, completely self-contained, automatically controlled, water atomizer, air distributor designed for minimum maintenance and very low operating coast.

Generally used in Textile, Tea, Tobacco Cashew nut industries etc



PNEUMATIC CONVEYING SYSTEMS

We can delivers advanced pneumatic conveying systems engineered for efficient, dust-free transport and separation of dry bulk materials such as powders and granules. Utilizing both positive pressure and vacuum conveying technologies, these systems move materials swiftly and safely through enclosed pipelines, minimizing product loss and contamination. Their modular, easy-to-install design ensures seamless integration into existing operations across industries like food, pharmaceuticals, and plastics, while low- maintenance components and energy-efficient operation provide long-term reliability and cost savings

